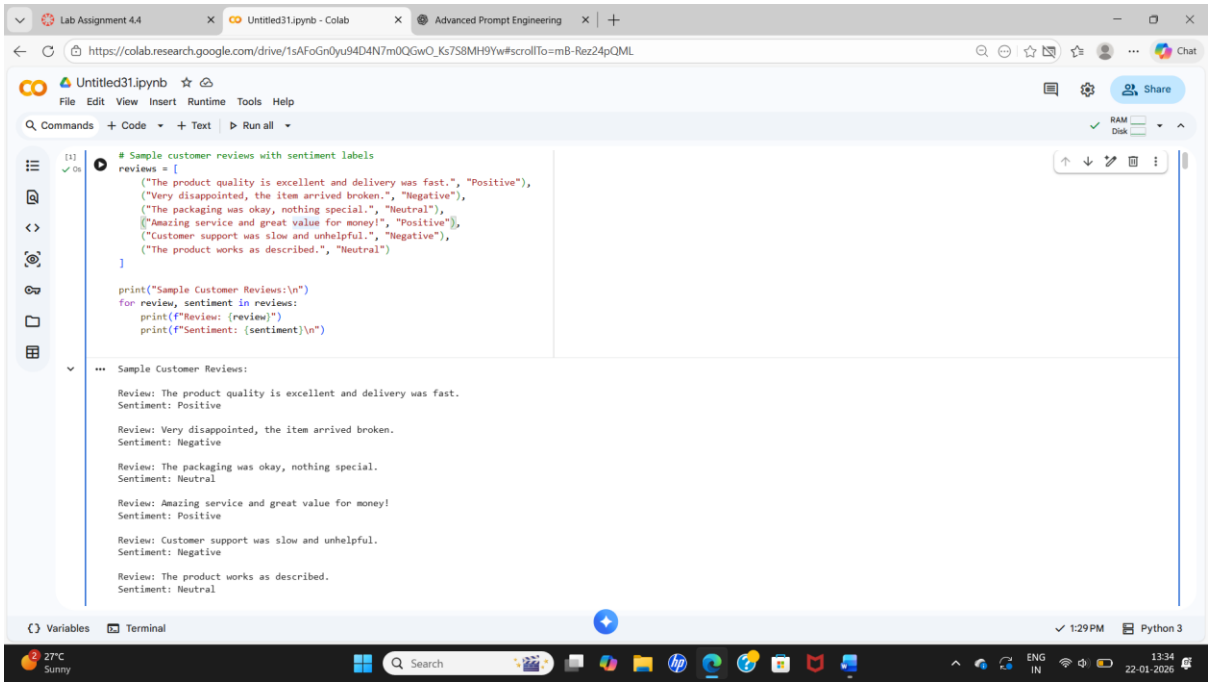
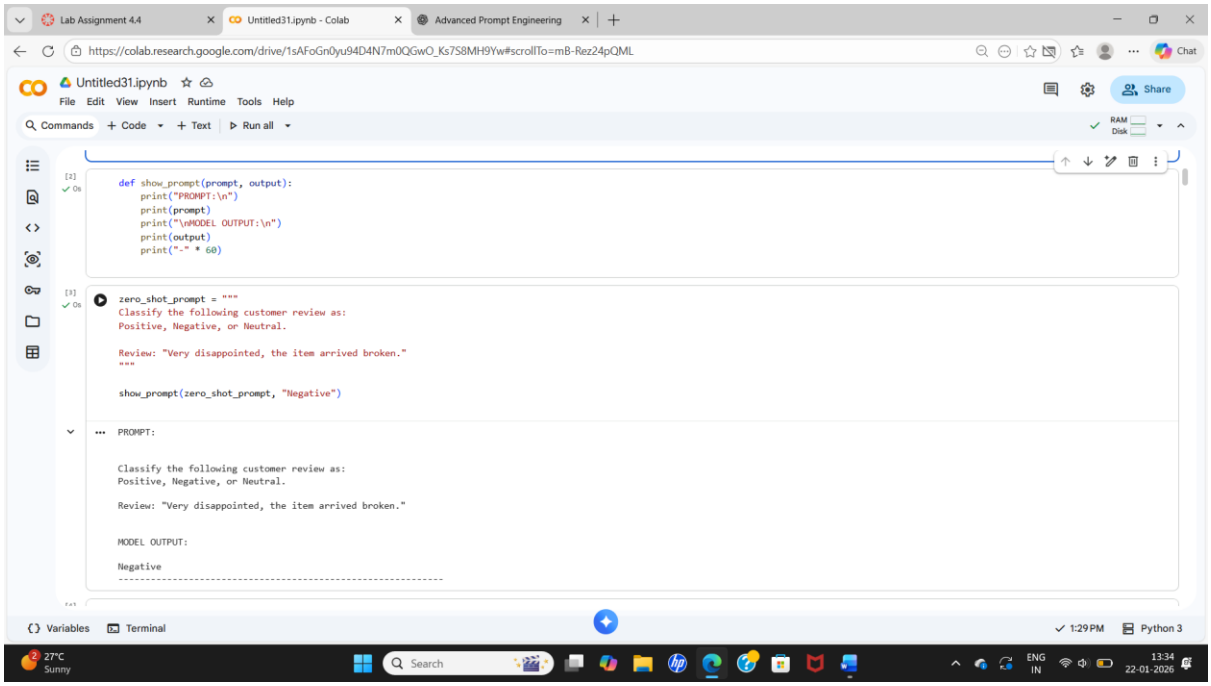
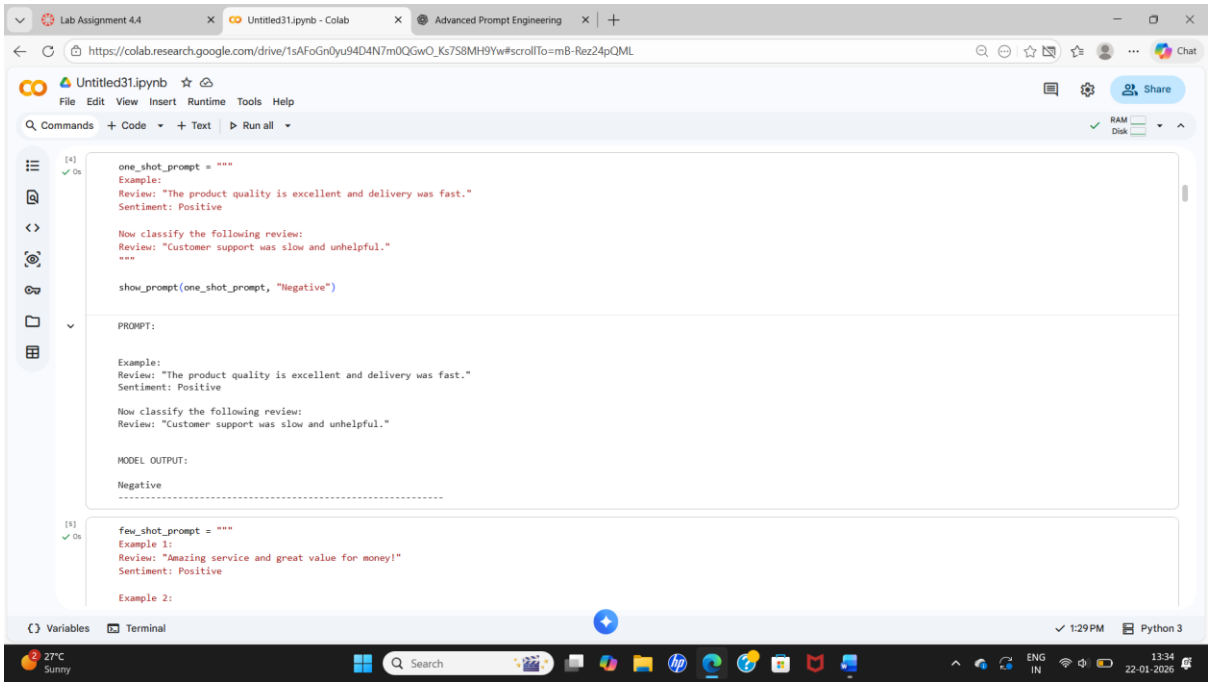
LAB ASSIGNMENT 4.4

NAME: R. PRANITHA REDDY ID NO: 2303A52248 SUBJECT: AI ASST CODING

1.Scenario



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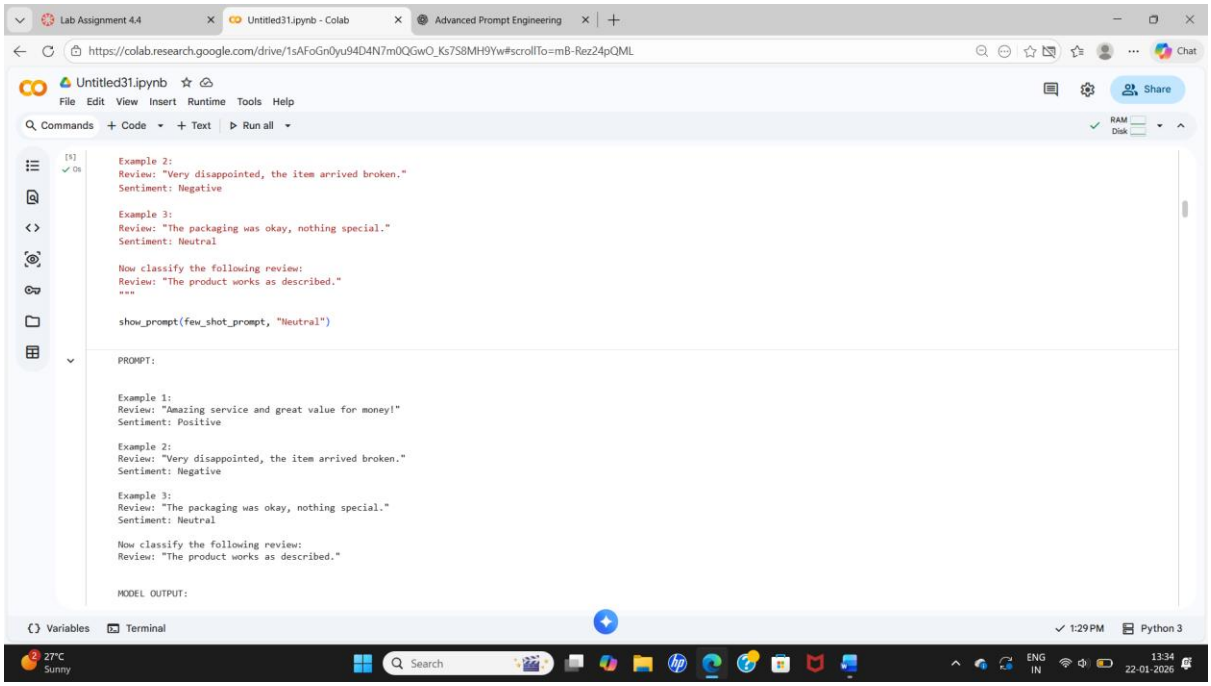
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
one_shot_prompt = """
Example:
Review: "The product quality is excellent and delivery was fast."
Sentiment: Positive
Now classify the following review:
Review: "Customer support was slow and unhelpful."
"""
show_prompt(one_shot_prompt, "Negative")
PROMPT:
Example:
Review: "The product quality is excellent and delivery was fast."
Sentiment: Positive
Now classify the following review:
Review: "Customer support was slow and unhelpful."
MODEL OUTPUT:
Negative
few_shot_prompt = """
Example 1:
Review: "Amazing service and great value for money!"
Sentiment: Positive
Example 2:
Variables Terminal
1:29 PM Python 3
27°C Sunny
Search
ENG IN
13:34 22-01-2026


Lab Assignment 4.4
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Advanced Prompt Engineering
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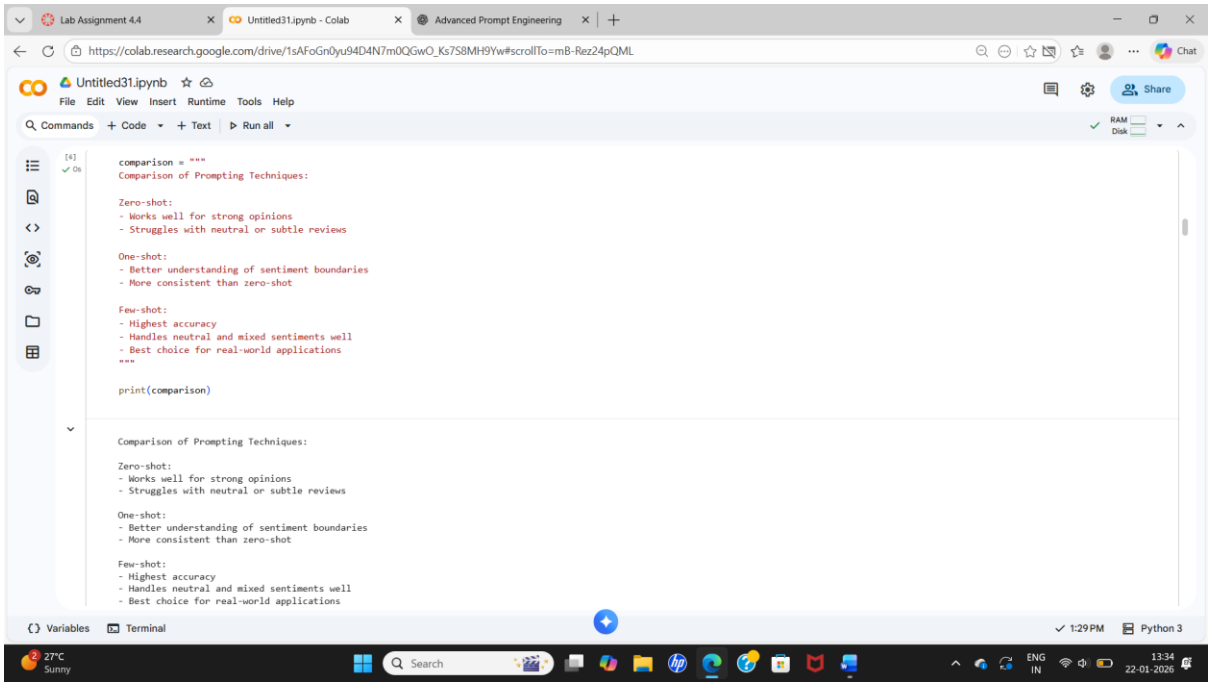
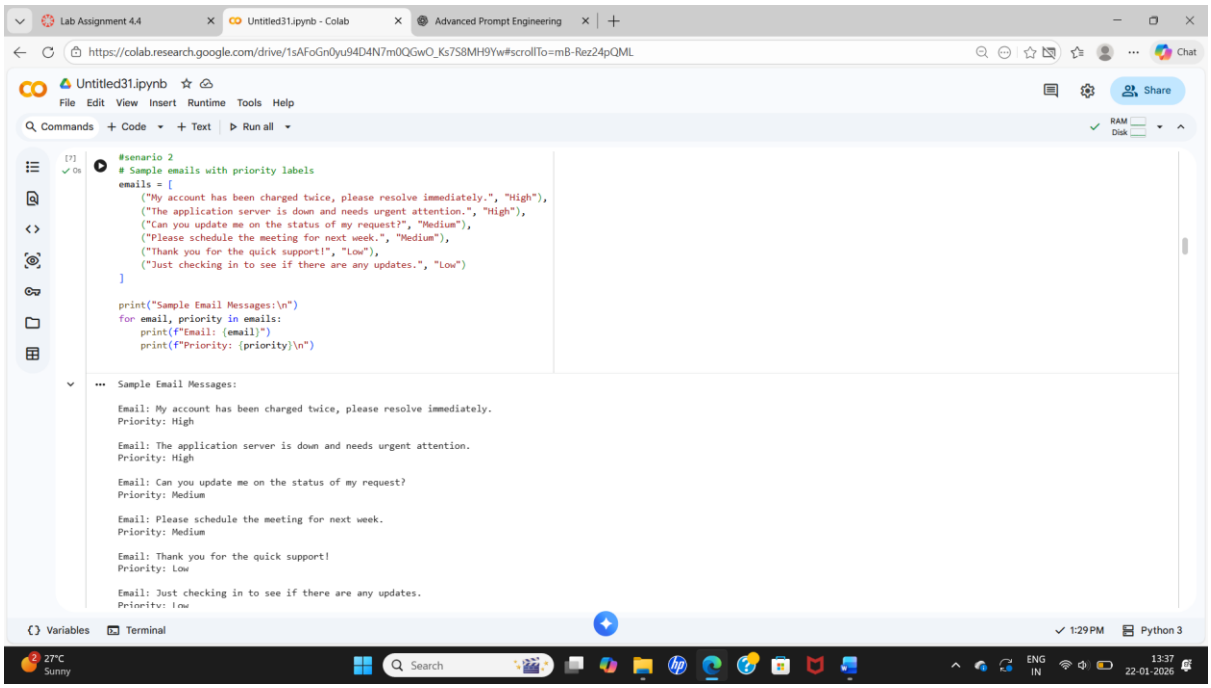
Share
Commands + Code + Text Run all
Example 2:
Review: "Very disappointed, the item arrived broken."
Sentiment: Negative
Example 3:
Review: "The packaging was okay, nothing special."
Sentiment: Neutral
Now classify the following review:
Review: "The product works as described."
"""
show_prompt(few_shot_prompt, "Neutral")
PROMPT:
Example 1:
Review: "Amazing service and great value for money!"
Sentiment: Positive
Example 2:
Review: "Very disappointed, the item arrived broken."
Sentiment: Negative
Example 3:
Review: "The packaging was okay, nothing special."
Sentiment: Neutral
Now classify the following review:
Review: "The product works as described."
MODEL OUTPUT:
Variables Terminal
1:29 PM Python 3
27°C Sunny
Search
ENG IN
13:34 22-01-2026



FINAL CONCLUSION:

Few-shot prompting provides the best sentiment classification accuracy because multiple examples help the model understand sentiment patterns, tone, and neutrality more effectively than zero-shot or one-shot methods.

2.Scenario


Lab Assignment 4.4
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Advanced Prompt Engineering
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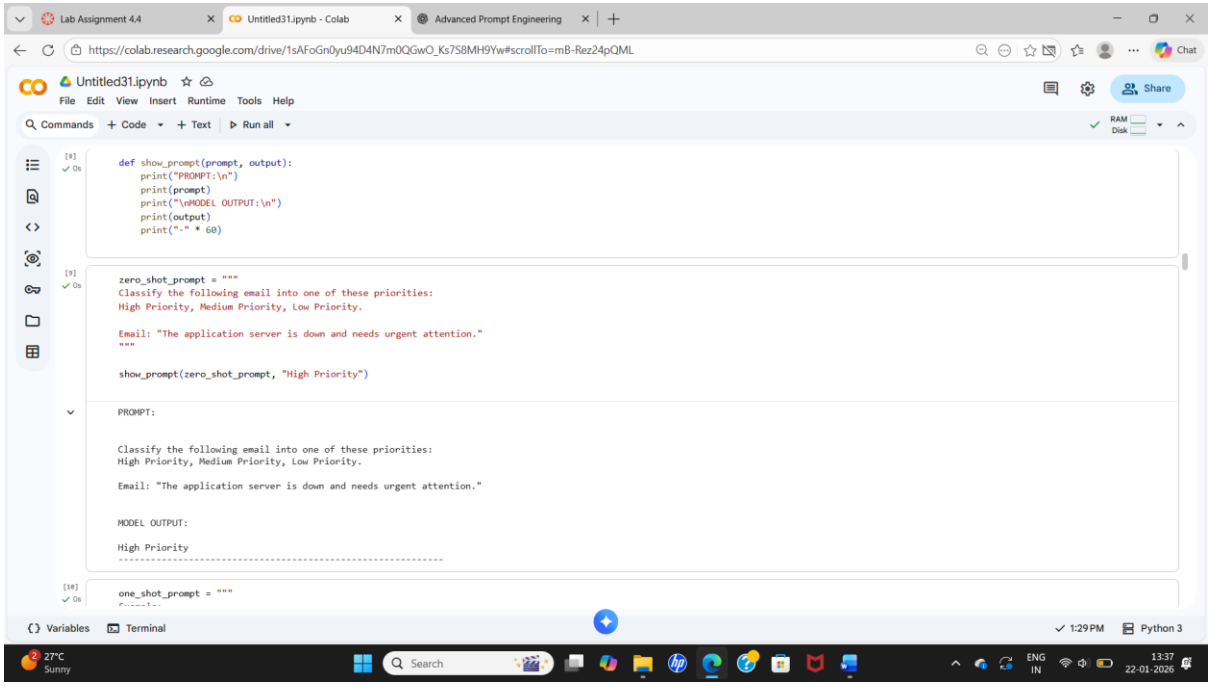
Untitled31.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Share
#senario 2
# Sample emails with priority labels
emails = [
("My account has been charged twice, please resolve immediately.", "High"),
("The application server is down and needs urgent attention.", "High"),
("Can you update me on the status of my request?", "Medium"),
("Please schedule the meeting for next week.", "Medium"),
("Thank you for the quick support!", "Low"),
("Just checking in to see if there are any updates.", "Low")
]
print("Sample Email Messages:\n")
for email, priority in emails:
print(f"Email: {email}")
print(f"Priority: {priority}\n")
Sample Email Messages:
Email: My account has been charged twice, please resolve immediately.
Priority: High
Email: The application server is down and needs urgent attention.
Priority: High
Email: Can you update me on the status of my request?
Priority: Medium
Email: Please schedule the meeting for next week.
Priority: Medium
Email: Thank you for the quick support!
Priority: Low
Email: Just checking in to see if there are any updates.
Variables Terminal
1:29 PM Python 3
27°C Sunny
Search
ENG IN
13:37 22-01-2026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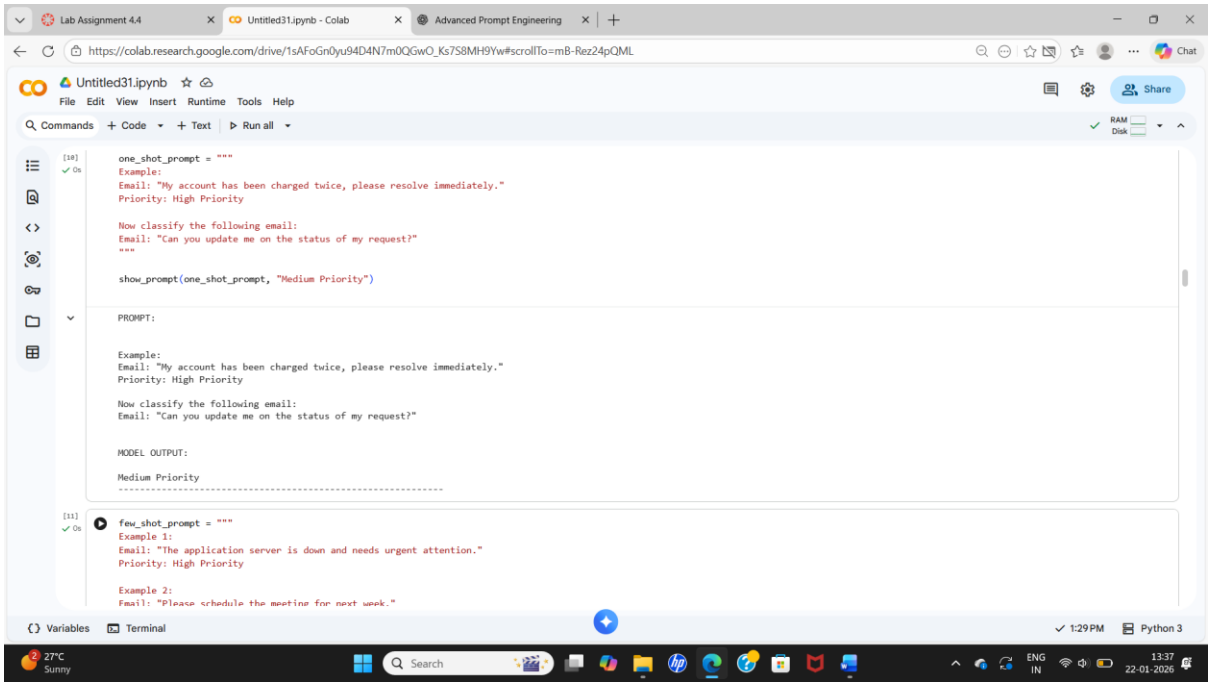
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Share
def show_prompt(prompt, output):
print("PROMPT:\n")
print(prompt)
print("\nMODEL OUTPUT:\n")
print(output)
print("-" * 60)
zero_shot_prompt = """
Classify the following email into one of these priorities:
High Priority, Medium Priority, Low Priority.
Email: "The application server is down and needs urgent attention."
"""
show_prompt(zero_shot_prompt, "High Priority")
PROMPT:
Classify the following email into one of these priorities:
High Priority, Medium Priority, Low Priority.
Email: "The application server is down and needs urgent attention."
MODEL OUTPUT:
High Priority
one_shot_prompt = """
Variables Terminal
1:29 PM Python 3
27°C Sunny
Search
ENG IN
13:37 22-01-2026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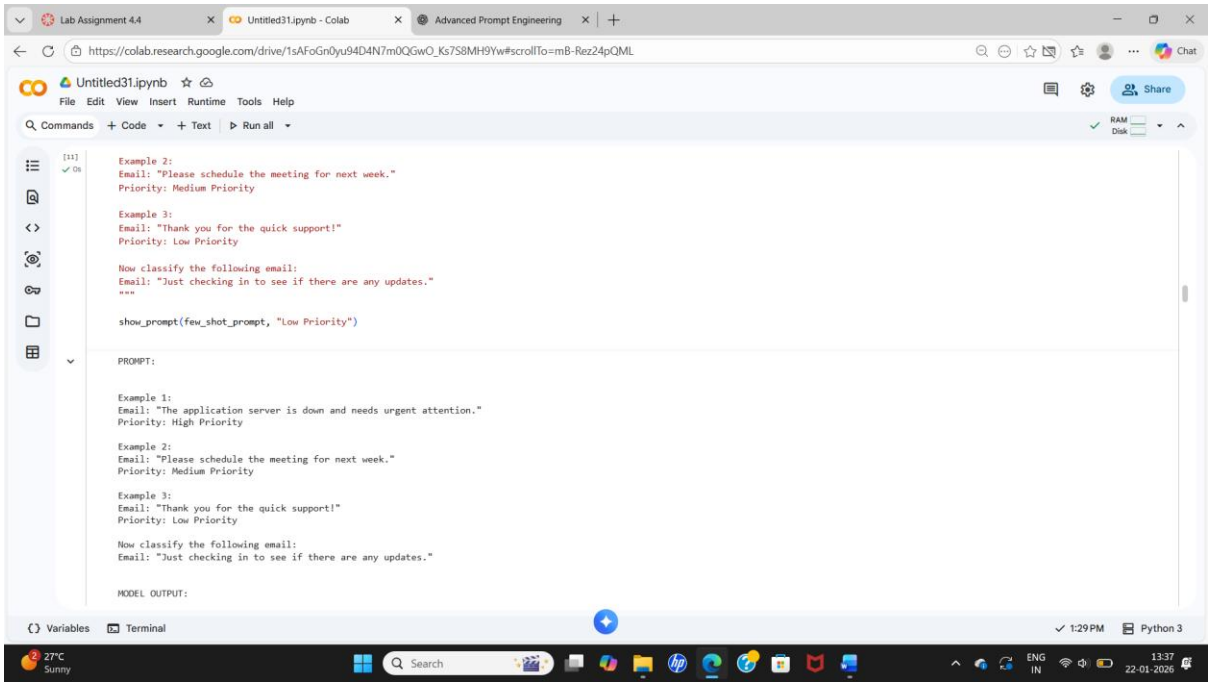
Untitled31.ipynb
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
one_shot_prompt = """
Example:
Email: "My account has been charged twice, please resolve immediately."
Priority: High Priority
Now classify the following email:
Email: "Can you update me on the status of my request?"
"""
show_prompt(one_shot_prompt, "Medium Priority")
PROMPT:
Example:
Email: "My account has been charged twice, please resolve immediately."
Priority: High Priority
Now classify the following email:
Email: "Can you update me on the status of my request?"
MODEL OUTPUT:
Medium Priority
few_shot_prompt = """
Example 1:
Email: "The application server is down and needs urgent attention."
Priority: High Priority
Example 2:
Variables Terminal
1:29 PM Python 3
27°C Sunny
Search
ENG IN
13:37 22-01-2026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Advanced Prompt Engineering
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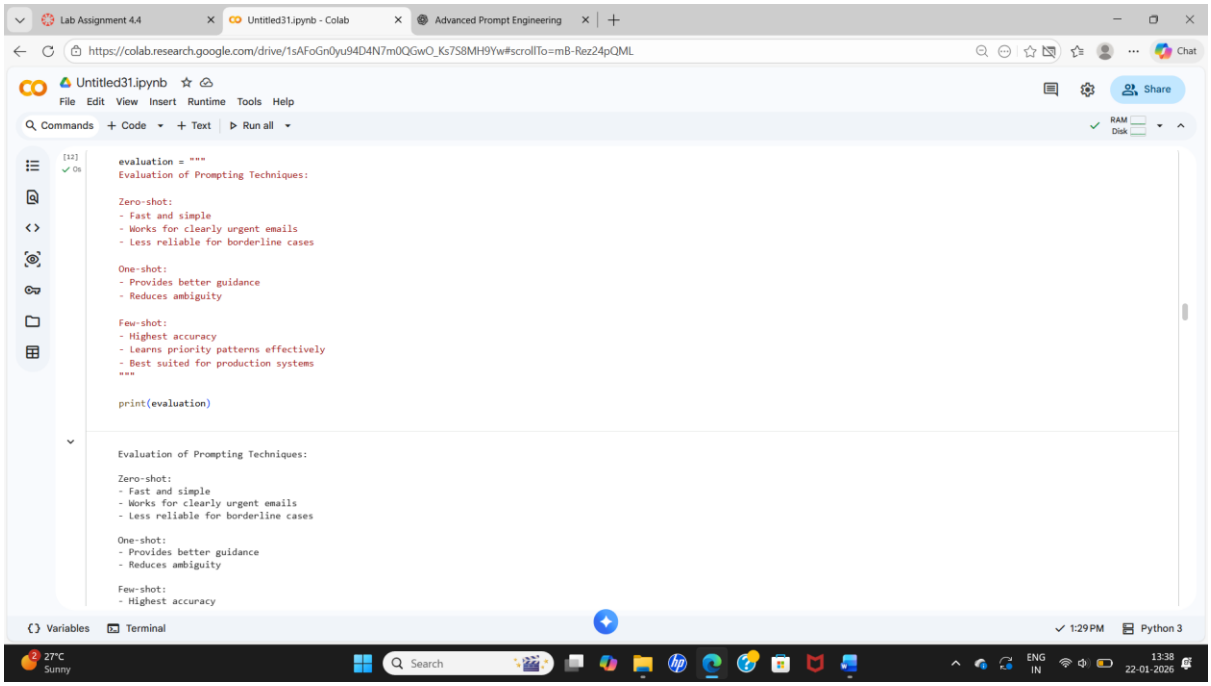
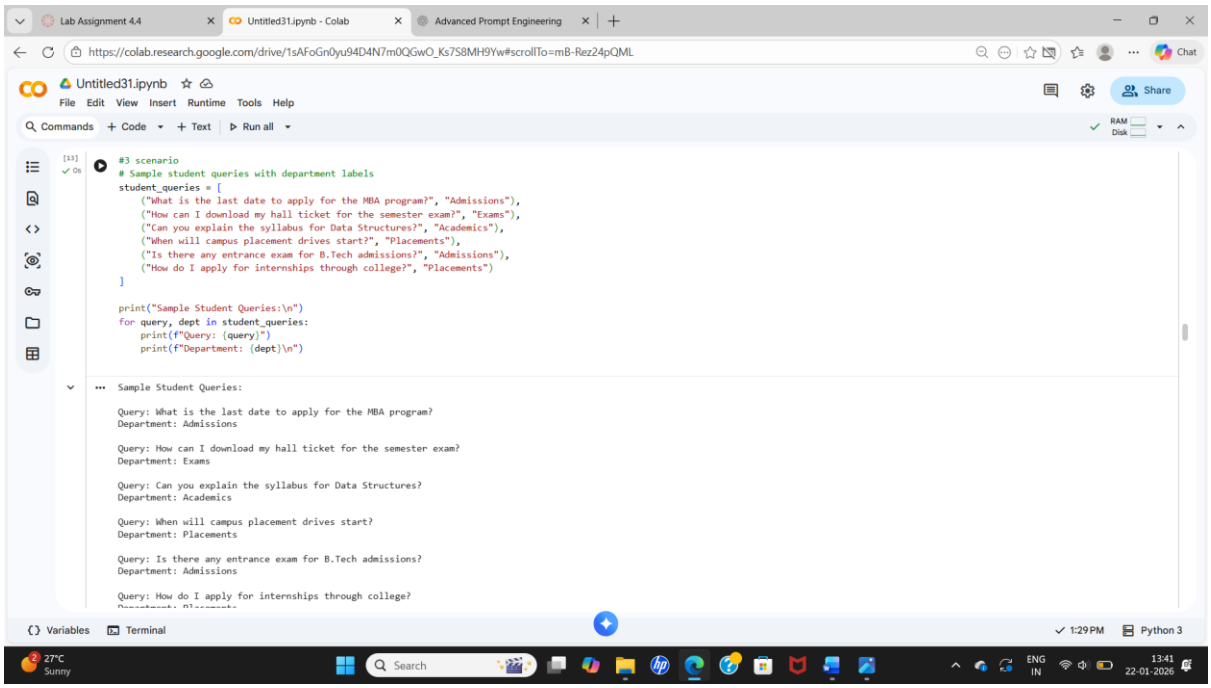
Share
Commands + Code + Text Run all
Example 2:
Email: "Please schedule the meeting for next week."
Priority: Medium Priority
Example 3:
Email: "Thank you for the quick support!"
Priority: Low Priority
Now classify the following email:
Email: "Just checking in to see if there are any updates."
"""
show_prompt(few_shot_prompt, "Low Priority")
PROMPT:
Example 1:
Email: "The application server is down and needs urgent attention."
Priority: High Priority
Example 2:
Email: "Please schedule the meeting for next week."
Priority: Medium Priority
Example 3:
Email: "Thank you for the quick support!"
Priority: Low Priority
Now classify the following email:
Email: "Just checking in to see if there are any updates."
MODEL OUTPUT:
Variables Terminal
1:29 PM Python 3
27°C Sunny
Search
ENG IN
13:37 22-01-2026



FINAL CONCLUSION:

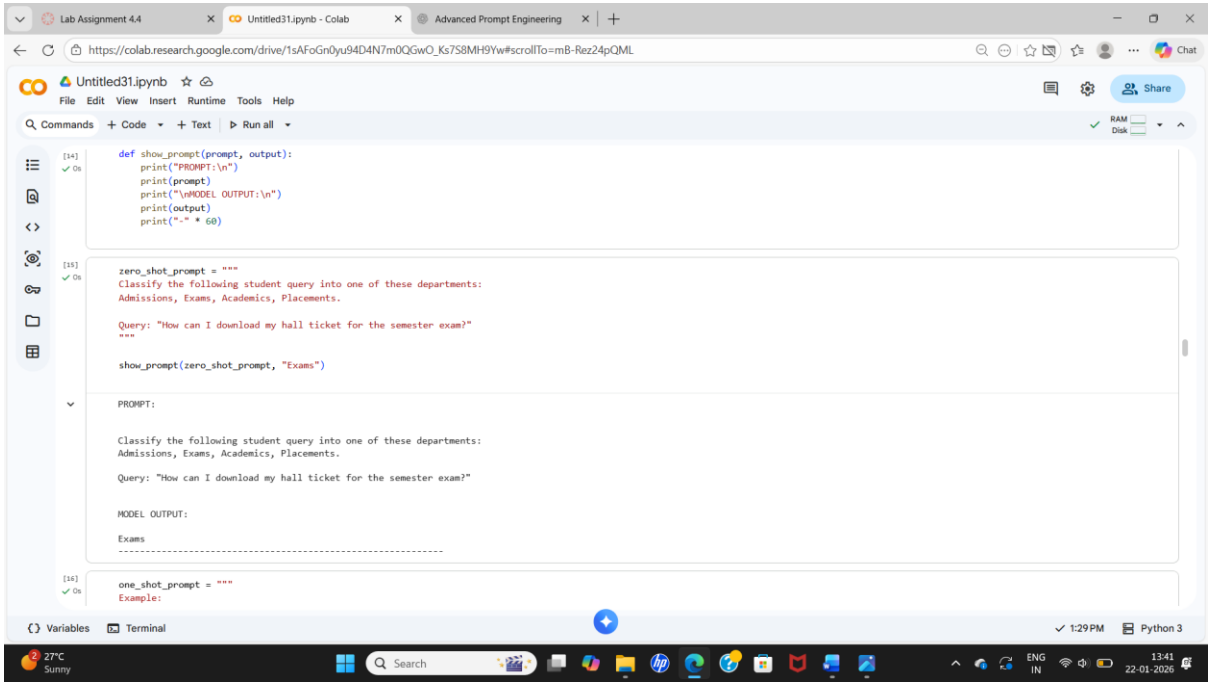
Few-shot prompting produces the most reliable results because multiple examples help the model clearly distinguish between high, medium, and low urgency patterns in emails.

3.Scenario



```
#3 scenario
# Sample student queries with department labels
student_queries = [
    ("What is the last date to apply for the MBA program?", "Admissions"),
    ("How can I download my hall ticket for the semester exam?", "Exams"),
    ("Can you explain the syllabus for Data Structures?", "Academics"),
    ("When will campus placement drives start?", "Placements"),
    ("Is there any entrance exam for B.Tech admissions?", "Admissions"),
    ("How do I apply for internships through college?", "Placements")
]

print("Sample Student Queries:\n")
for query, dept in student_queries:
    print(f"Query: {query}")
    print(f"Department: {dept}\n")
```

```
Sample Student Queries:

Query: What is the last date to apply for the MBA program?
Department: Admissions

Query: How can I download my hall ticket for the semester exam?
Department: Exams

Query: Can you explain the syllabus for Data Structures?
Department: Academics

Query: When will campus placement drives start?
Department: Placements

Query: Is there any entrance exam for B.Tech admissions?
Department: Admissions

Query: How do I apply for internships through college?
```

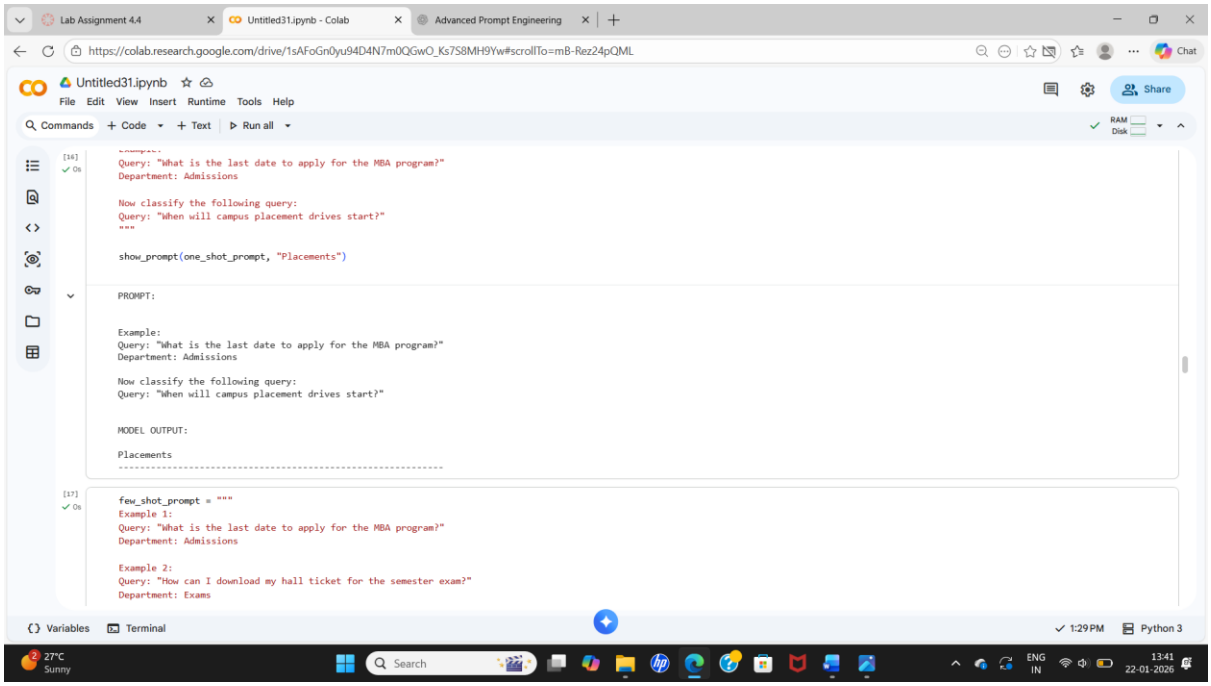


```
def show_prompt(prompt, output):
    print("PROMPT:\n")
    print(prompt)
    print("\nMODEL OUTPUT:\n")
    print(output)
    print("-" * 60)
```

```
zero_shot_prompt = """
Classify the following student query into one of these departments:
Admissions, Exams, Academics, Placements.

Query: "How can I download my hall ticket for the semester exam?"
"""

show_prompt(zero_shot_prompt, "Exams")
```

```
PROMPT:


Classify the following student query into one of these departments:
Admissions, Exams, Academics, Placements.

Query: "How can I download my hall ticket for the semester exam?"


MODEL OUTPUT:

Exams
------------------------------------------------------------
```

```
one_shot_prompt = """
Example:
```


Lab Assignment 4.4
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Advanced Prompt Engineering
https://colab.research.google.com/drive/1sAFoGn0yu94D4N7m0QGwO_Ks7S8MH9Yw#scrollTo=mB-Rez24pQML
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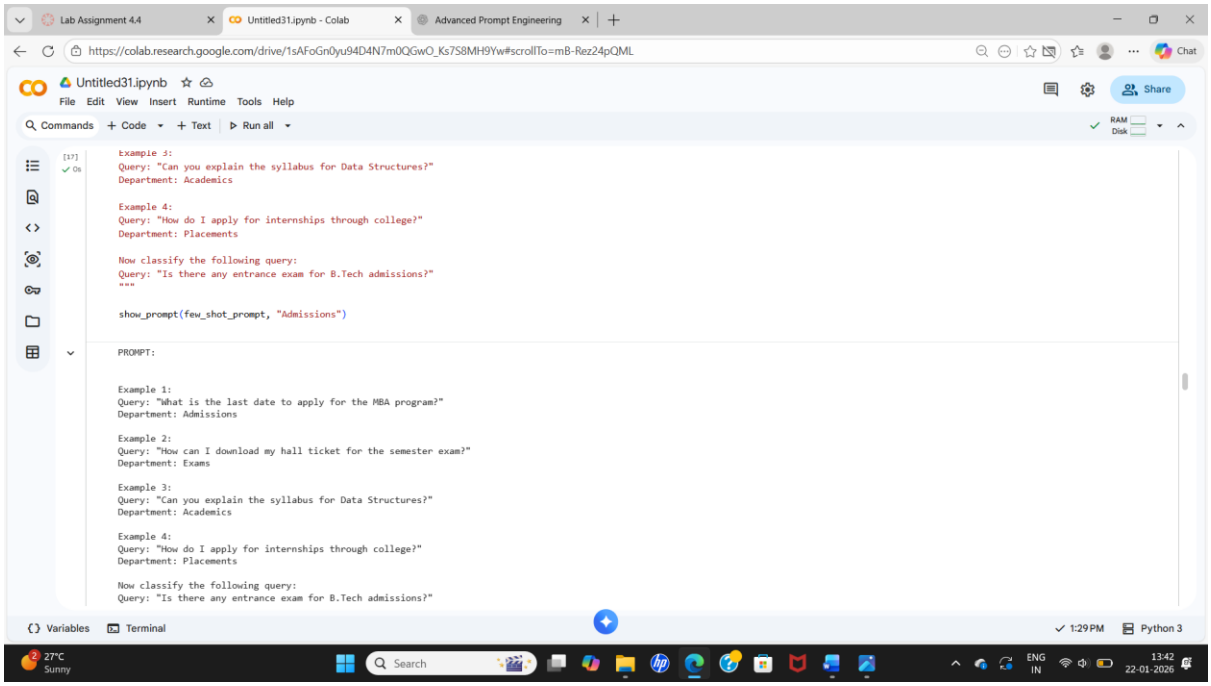
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
Query: "What is the last date to apply for the MBA program?"
Department: Admissions
Now classify the following query:
Query: "When will campus placement drives start?"
"""
show_prompt(one_shot_prompt, "Placements")
PROMPT:
Example:
Query: "What is the last date to apply for the MBA program?"
Department: Admissions
Now classify the following query:
Query: "When will campus placement drives start?"
MODEL OUTPUT:
Placements
few_shot_prompt = """
Example 1:
Query: "What is the last date to apply for the MBA program?"
Department: Admissions
Example 2:
Query: "How can I download my hall ticket for the semester exam?"
Department: Exams
Variables Terminal
1:29 PM Python 3
27°C Sunny
Search
ENG IN
13:41 22-01-2026


Lab Assignment 4.4
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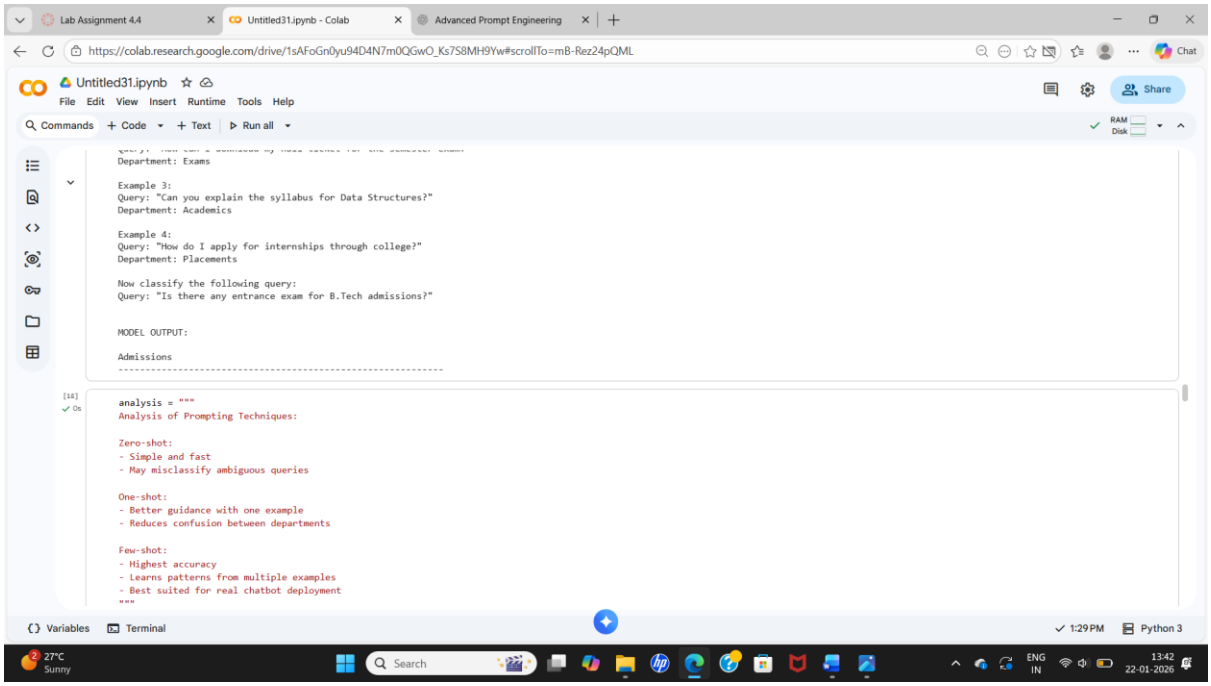
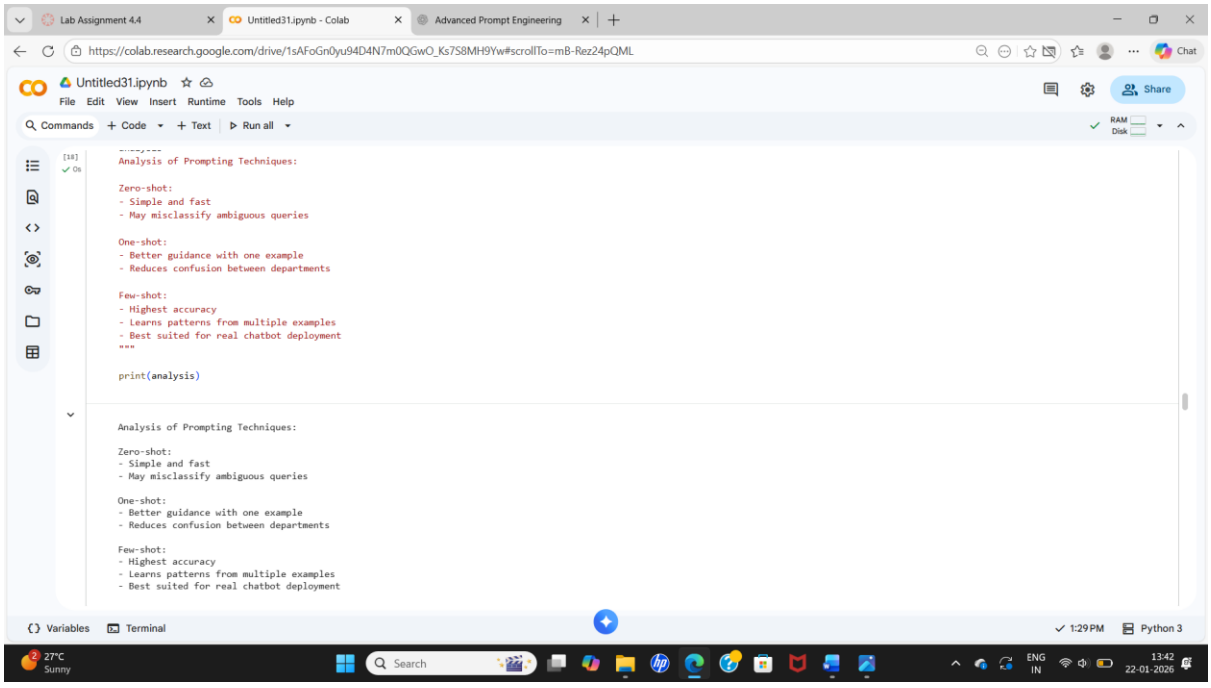
Advanced Prompt Engineering
https://colab.research.google.com/drive/1sAFoGn0yu94D4N7m0QGwO_Ks7S8MH9Yw#scrollTo=mB-Rez24pQML
Untitled31.ipynb
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
Example 3:
Query: "Can you explain the syllabus for Data Structures?"
Department: Academics
Example 4:
Query: "How do I apply for internships through college?"
Department: Placements
Now classify the following query:
Query: "Is there any entrance exam for B.Tech admissions?"
"""
show_prompt(few_shot_prompt, "Admissions")
PROMPT:
Example 1:
Query: "What is the last date to apply for the MBA program?"
Department: Admissions
Example 2:
Query: "How can I download my hall ticket for the semester exam?"
Department: Exams
Example 3:
Query: "Can you explain the syllabus for Data Structures?"
Department: Academics
Example 4:
Query: "How do I apply for internships through college?"
Department: Placements
Now classify the following query:
Query: "Is there any entrance exam for B.Tech admissions?"
Variables Terminal
1:29 PM Python 3
27°C Sunny
Search
ENG IN
13:42 22-01-2026

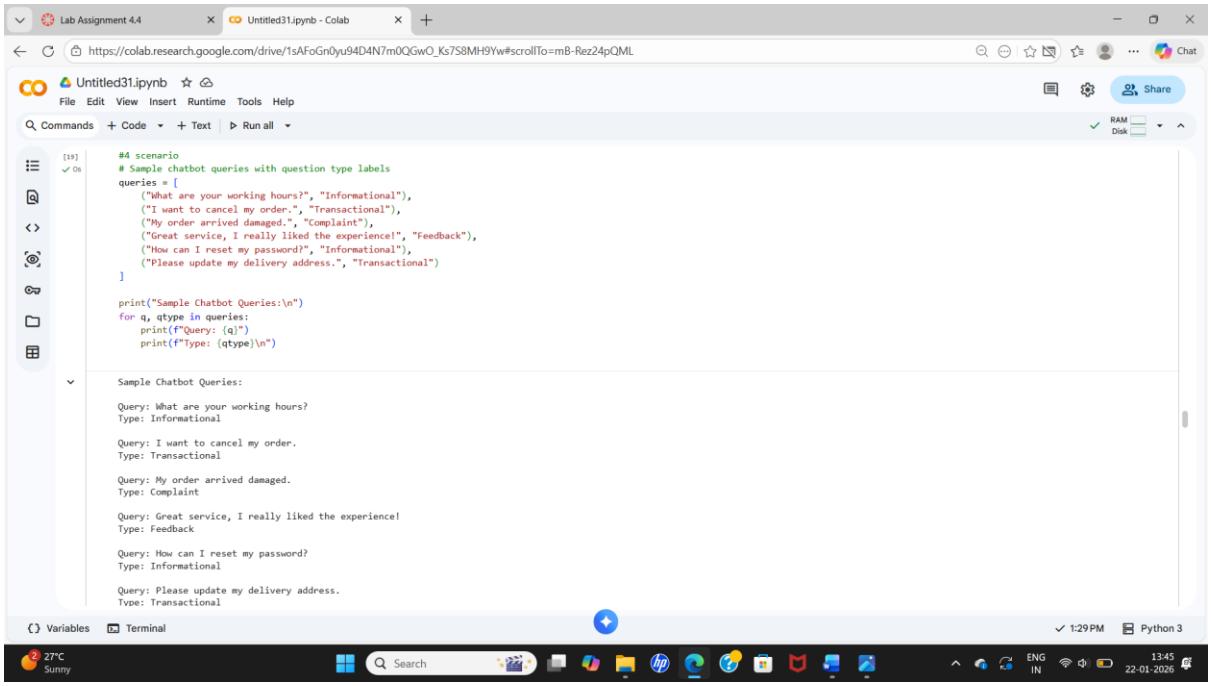




FINAL CONCLUSION:

Few-shot prompting significantly improves student query routing accuracy by providing contextual patterns, making it the most effective approach for university chatbots.

4.Scenario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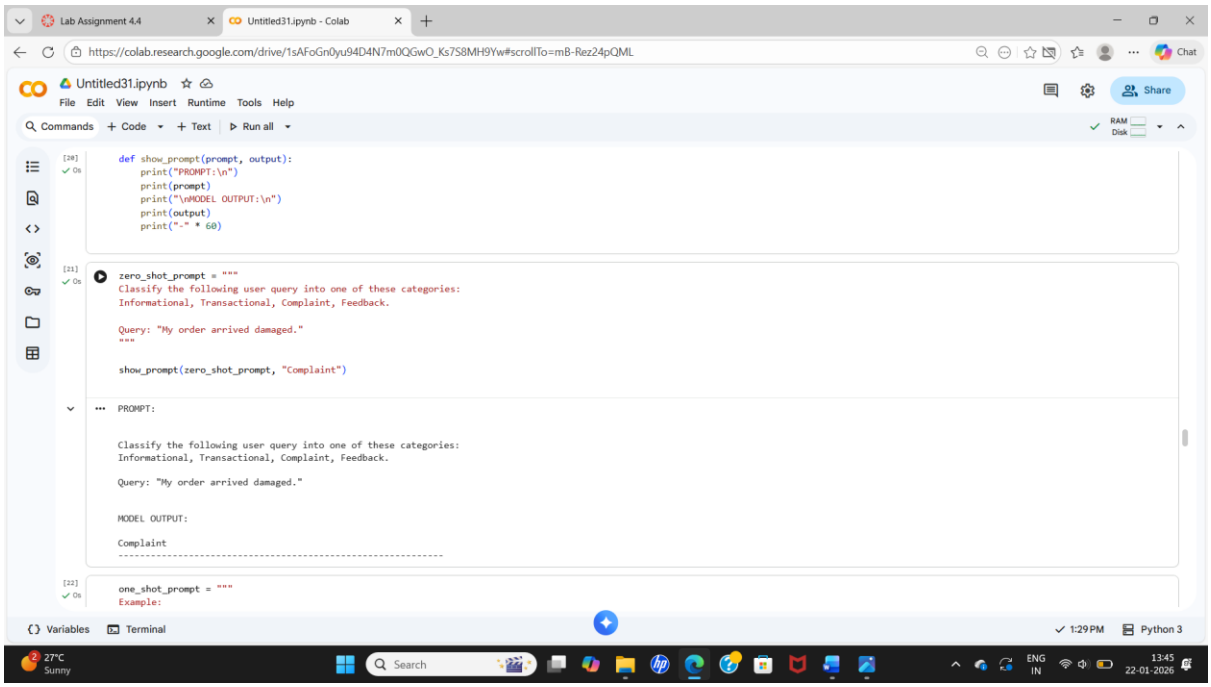
Commands + Code + Text Run all
#4 scenario
# Sample chatbot queries with question type labels
queries = [
("What are your working hours?", "Informational"),
("I want to cancel my order.", "Transactional"),
("My order arrived damaged.", "Complaint"),
("Great service, I really liked the experience!", "Feedback"),
("How can I reset my password?", "Informational"),
("Please update my delivery address.", "Transactional")
]
print("Sample Chatbot Queries:\n")
for q, qtype in queries:
print(f"Query: {q}")
print(f"Type: {qtype}\n")
Sample Chatbot Queries:
Query: What are your working hours?
Type: Informational
Query: I want to cancel my order.
Type: Transactional
Query: My order arrived damaged.
Type: Complaint
Query: Great service, I really liked the experience!
Type: Feedback
Query: How can I reset my password?
Type: Informational
Query: Please update my delivery address.
Type: Transactional
Variables Terminal
1:29 PM Python 3
27°C Sunny
Search
ENG IN
13:45 22-01-2026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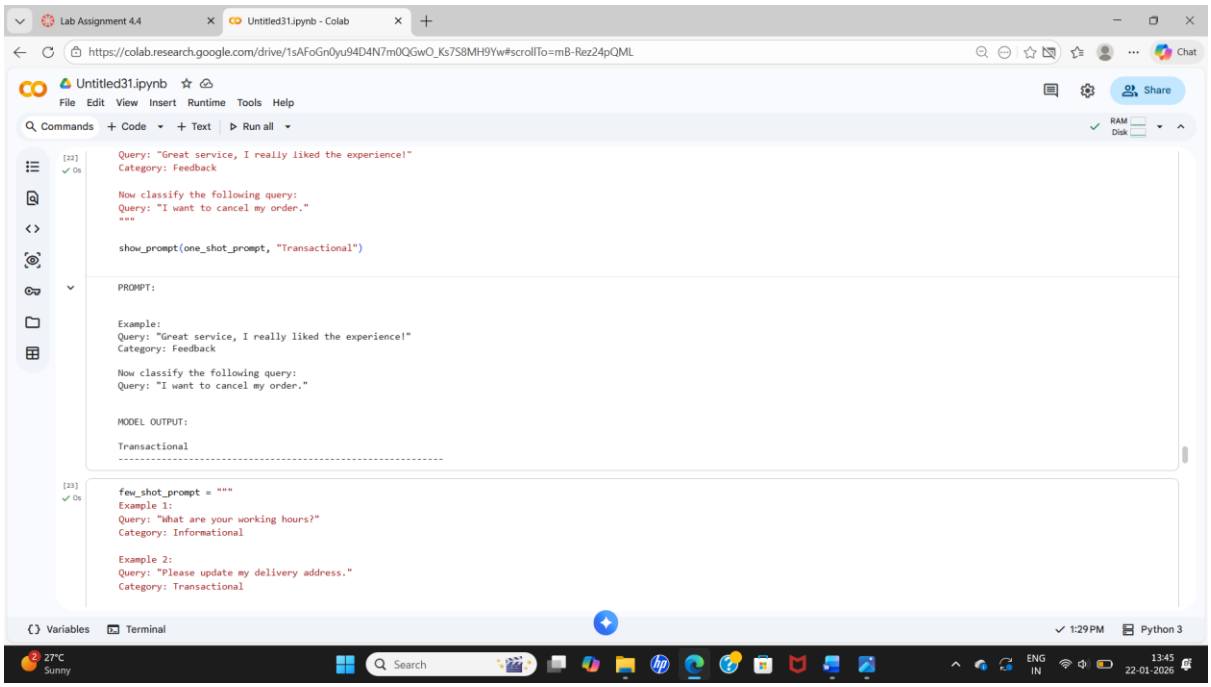
Untitled31.ipynb
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
def show_prompt(prompt, output):
print("PROMPT:\n")
print(prompt)
print("\nMODEL OUTPUT:\n")
print(output)
print("-" * 60)
zero_shot_prompt = """
Classify the following user query into one of these categories:
Informational, Transactional, Complaint, Feedback.
Query: "My order arrived damaged."
"""
show_prompt(zero_shot_prompt, "Complaint")
PROMPT:
Classify the following user query into one of these categories:
Informational, Transactional, Complaint, Feedback.
Query: "My order arrived damaged."
MODEL OUTPUT:
Complaint
one_shot_prompt = """
Example:
Variables Terminal
1:29 PM Python 3
27°C Sunny
Search
ENG IN
13:45 22-01-2026


Lab Assignment 4.4
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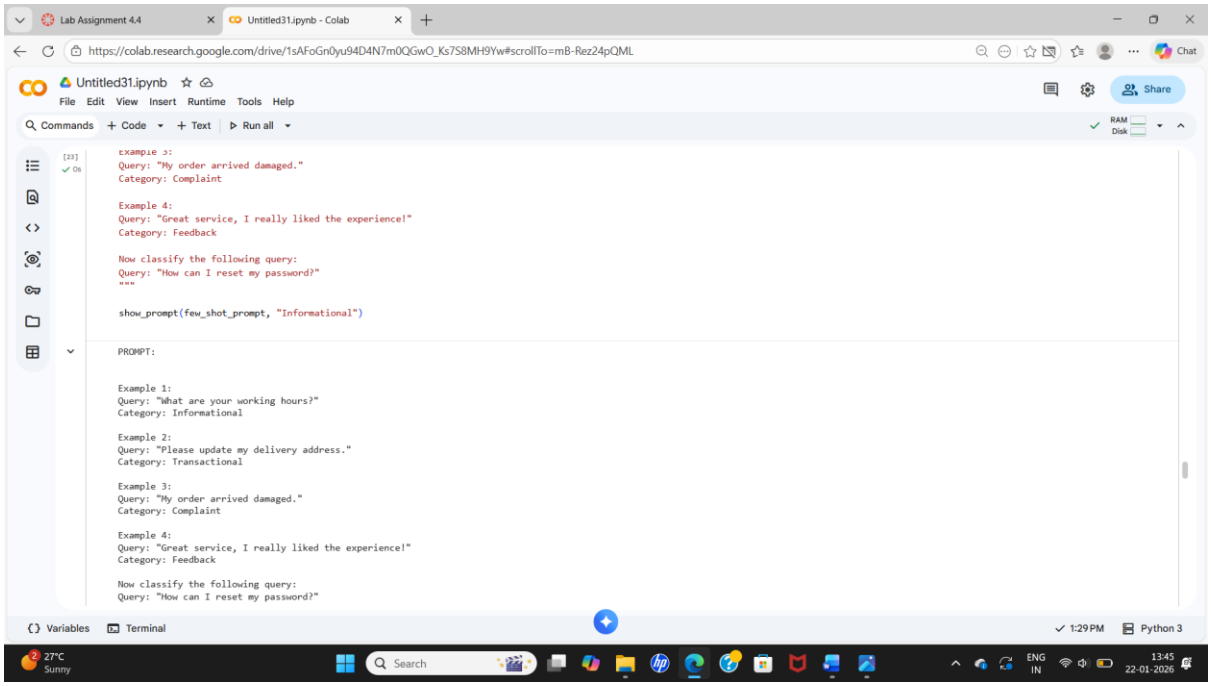
Untitled31.ipynb
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
RAM Disk
[22] 0s
Query: "Great service, I really liked the experience!"
Category: Feedback
Now classify the following query:
Query: "I want to cancel my order."
"""
show_prompt(one_shot_prompt, "Transactional")
PROMPT:
Example:
Query: "Great service, I really liked the experience!"
Category: Feedback
Now classify the following query:
Query: "I want to cancel my order."
MODEL OUTPUT:
Transactional
[23] 0s
few_shot_prompt = """
Example 1:
Query: "What are your working hours?"
Category: Informational
Example 2:
Query: "Please update my delivery address."
Category: Transactional
Variables Terminal
1:29 PM Python 3
27°C Sunny
Search
ENG IN
13:45 22-01-2026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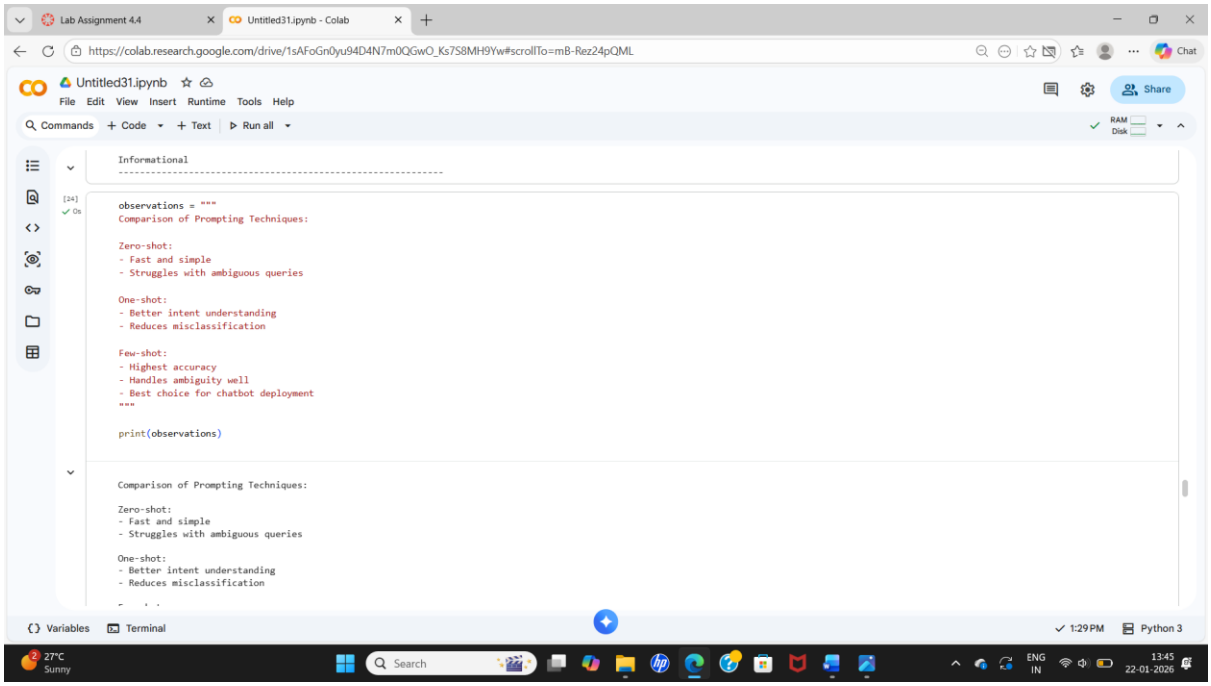
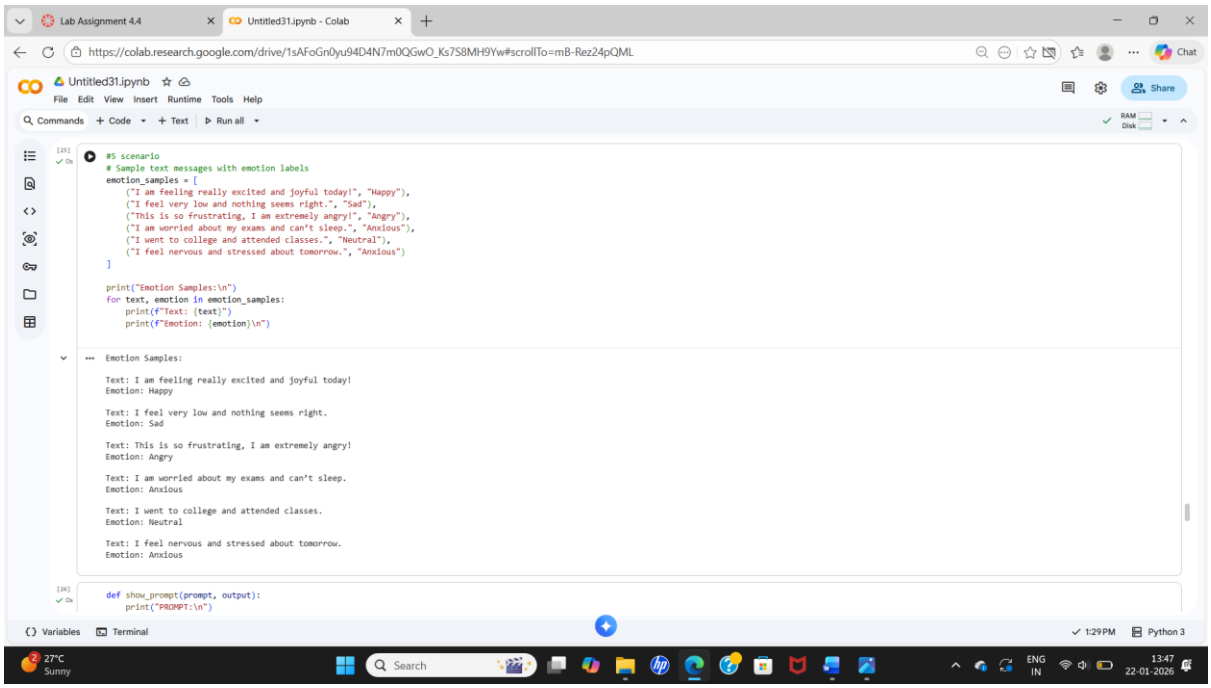
Commands + Code + Text Run all
RAM Disk
[23] 0s
Example 3:
Query: "My order arrived damaged."
Category: Complaint
Example 4:
Query: "Great service, I really liked the experience!"
Category: Feedback
Now classify the following query:
Query: "How can I reset my password?"
"""
show_prompt(few_shot_prompt, "Informational")
PROMPT:
Example 1:
Query: "What are your working hours?"
Category: Informational
Example 2:
Query: "Please update my delivery address."
Category: Transactional
Example 3:
Query: "My order arrived damaged."
Category: Complaint
Example 4:
Query: "Great service, I really liked the experience!"
Category: Feedback
Now classify the following query:
Query: "How can I reset my password?"
Variables Terminal
1:29 PM Python 3
27°C Sunny
Search
ENG IN
13:45 22-01-2026



FINAL CONCLUSION:

Few-shot prompting significantly improves chatbot question type detection by providing contextual examples, making it more robust and reliable than zero-shot and one-shot methods.

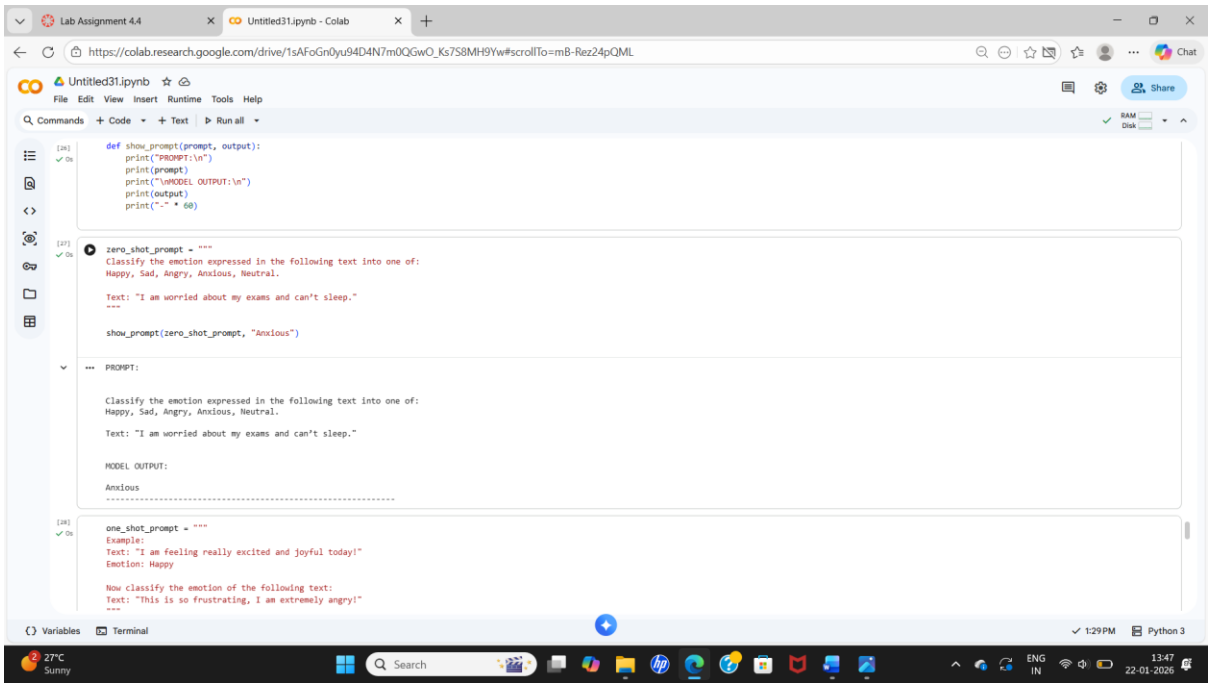
5.Scenario



```
#5 scenario
# Sample text messages with emotion labels
emotion_samples = [
    ("I am feeling really excited and joyful today!", "Happy"),
    ("I feel very low and nothing seems right.", "Sad"),
    ("This is so frustrating, I am extremely angry!", "Angry"),
    ("I am worried about my exams and can't sleep.", "Anxious"),
    ("I went to college and attended classes.", "Neutral"),
    ("I feel nervous and stressed about tomorrow.", "Anxious")
]

print("Emotion Samples:\n")
for text, emotion in emotion_samples:
    print(f"Text: {text}")
    print(f"Emotion: {emotion}\n")
```

```
Emotion Samples:

Text: I am feeling really excited and joyful today!
Emotion: Happy

Text: I feel very low and nothing seems right.
Emotion: Sad

Text: This is so frustrating, I am extremely angry!
Emotion: Angry

Text: I am worried about my exams and can't sleep.
Emotion: Anxious

Text: I went to college and attended classes.
Emotion: Neutral

Text: I feel nervous and stressed about tomorrow.
Emotion: Anxious
```

```
def show_prompt(prompt, output):
    print("PROMPT:\n")
```

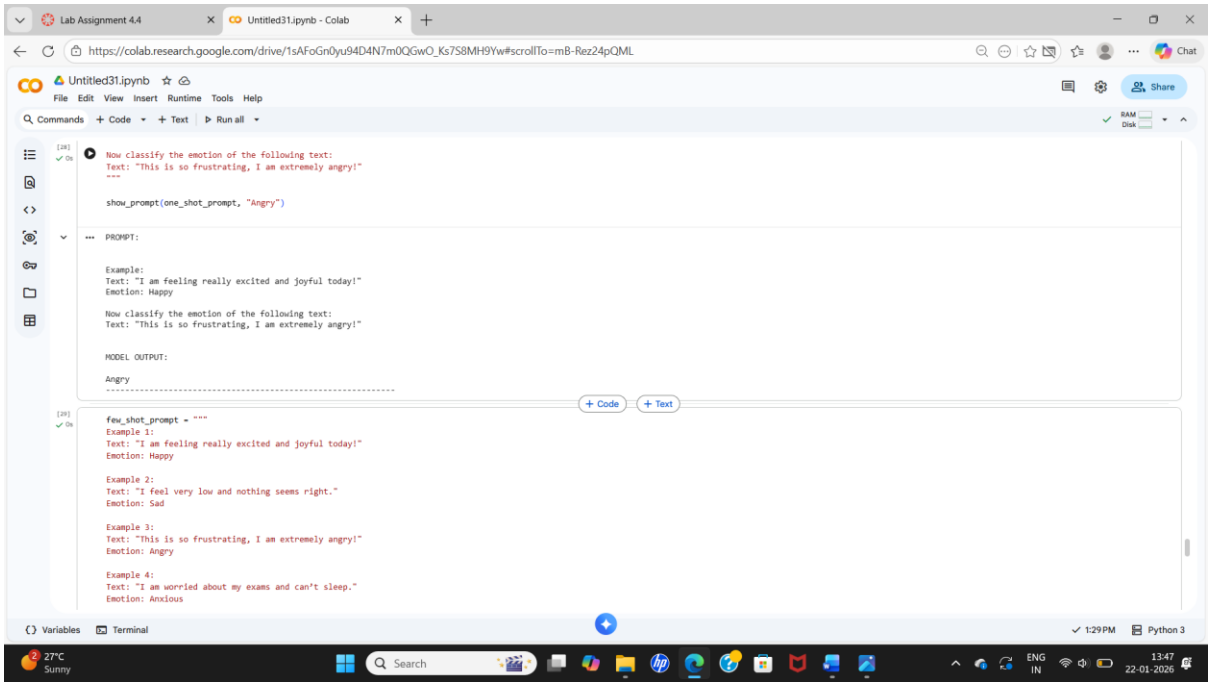


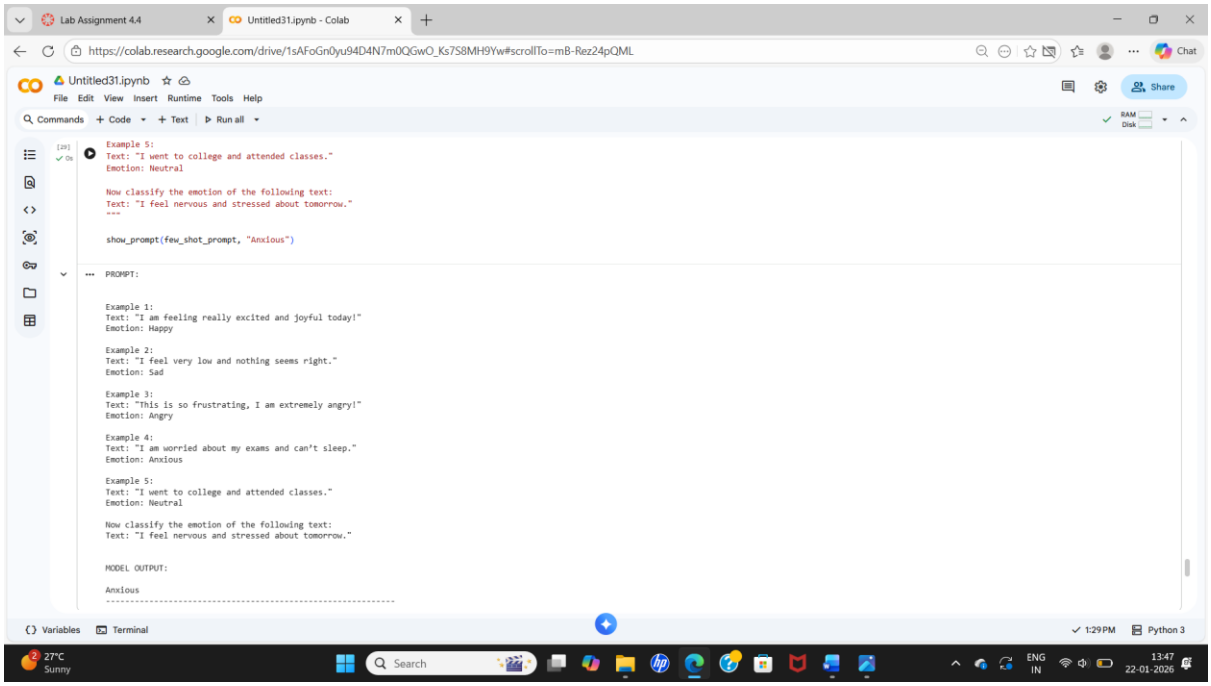
```
def show_prompt(prompt, output):
    print("PROMPT:\n")
    print(prompt)
    print("\nMODEL OUTPUT:\n")
    print(output)
    print("-" * 60)
```

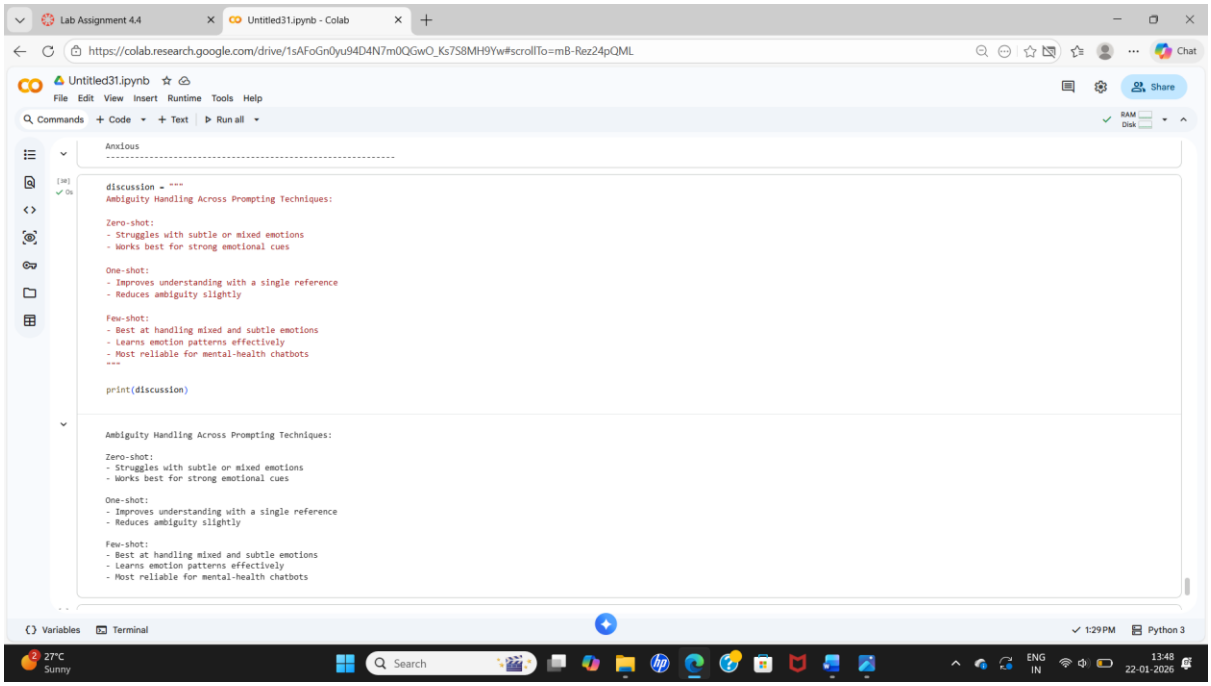
```
zero_shot_prompt = """
Classify the emotion expressed in the following text into one of:
Happy, Sad, Angry, Anxious, Neutral.

Text: "I am worried about my exams and can't sleep."
"""

show_prompt(zero_shot_prompt, "Anxious")
```

```
PROMPT:


Classify the emotion expressed in the following text into one of:
Happy, Sad, Angry, Anxious, Neutral.

Text: "I am worried about my exams and can't sleep."


MODEL OUTPUT:

Anxious
------------------------------------------------------------
```

```
one_shot_prompt = """
Example:
Text: "I am feeling really excited and joyful today!"
Emotion: Happy

Now classify the emotion of the following text:
Text: "This is so frustrating, I am extremely angry!"
"""
```


Lab Assignment 4.4
Untitled31.ipynb - Colab
https://colab.research.google.com/drive/1sAFoGn0yu94D4N7m0QGwO_Ks7S8MH9Yw#scrollTo=mB-Rez24pQML
Untitled31.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Share
Chat
Now classify the emotion of the following text:
Text: "This is so frustrating, I am extremely angry!"
"""
show_prompt(one_shot_prompt, "Angry")
PROMPT:
Example:
Text: "I am feeling really excited and joyful today!"
Emotion: Happy
Now classify the emotion of the following text:
Text: "This is so frustrating, I am extremely angry!"
MODEL OUTPUT:
Angry
+ Code + Text
few_shot_prompt = """
Example 1:
Text: "I am feeling really excited and joyful today!"
Emotion: Happy
Example 2:
Text: "I feel very low and nothing seems right."
Emotion: Sad
Example 3:
Text: "This is so frustrating, I am extremely angry!"
Emotion: Angry
Example 4:
Text: "I am worried about my exams and can't sleep."
Emotion: Anxious
Variables Terminal
1:29PM Python 3
27°C Sunny
Search
ENG IN
13:47 22-01-2026


Lab Assignment 4.4
Untitled31.ipynb - Colab
https://colab.research.google.com/drive/1sAFoGn0yu94D4N7m0QGwO_Ks7S8MH9Yw#scrollTo=mB-Rez24pQML
Untitled31.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Share
Chat
Example 5:
Text: "I went to college and attended classes."
Emotion: Neutral
Now classify the emotion of the following text:
Text: "I feel nervous and stressed about tomorrow."
"""
show_prompt(few_shot_prompt, "Anxious")
PROMPT:
Example 1:
Text: "I am feeling really excited and joyful today!"
Emotion: Happy
Example 2:
Text: "I feel very low and nothing seems right."
Emotion: Sad
Example 3:
Text: "This is so frustrating, I am extremely angry!"
Emotion: Angry
Example 4:
Text: "I am worried about my exams and can't sleep."
Emotion: Anxious
Example 5:
Text: "I went to college and attended classes."
Emotion: Neutral
Now classify the emotion of the following text:
Text: "I feel nervous and stressed about tomorrow."
MODEL OUTPUT:
Anxious
Variables Terminal
1:29PM Python 3
27°C Sunny
Search
ENG IN
13:47 22-01-2026



FINAL CONCLUSION:

Few-shot prompting provides the highest accuracy in emotion detection because multiple labelled examples help the model distinguish subtle emotional differences, making it ideal for mental-health applications.